

Isaac Mubiru

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PROFESSIONAL SUMMARY

Backend engineer with 6+ years designing and shipping **production backend systems** in **Node.js** and **TypeScript**. Led the zero-downtime migration of **20+ live applications** to Firebase/GCP, reporting directly to the CTO, and built the two-way MongoDB↔Firestore sync on **Google Pub/Sub** that kept both stacks consistent under parallel production traffic — designed for redelivery, replay and out-of-order messages. Architected **containerised microservices on Kubernetes** with Apache Kafka and gRPC, ran scheduled jobs over 2M+ records, and wrote the SDK documentation and Ops training that let a team operate a new platform without me. Comfortable owning ambiguous, product-shaped infrastructure problems end to end; daily user of AI-assisted engineering tooling.

CORE COMPETENCIES

Node.js & TypeScript

Production Backend Systems

Kubernetes & Containerised Services

Event-Driven Architecture (Kafka, Pub/Sub)

Zero-Downtime Migrations

GCP & AWS

Documentation & Developer Enablement

AI-Assisted Engineering Tooling

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote

- Built a **real-time two-way sync** between MongoDB and Firestore in **Node.js** on Google **Pub/Sub**, with idempotent, redelivery-safe message processing so both stores stayed consistent while serving live production traffic.
- Led the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Migrated all services off AWS (S3 → GCS via a Python pipeline), cutting infrastructure cost by **\$5,000/month** with no outage.
- Shipped Express Shipping (+\$40 revenue per order) and a 3D visualisation feature (video overlay + ffmpeg) that increased buyer conversion.

MTailor

Mar 2022 – Jul 2022

Backend Contractor · US Remote

- Built the migration pipeline that read from MongoDB, applied complex transformation logic, and wrote to Firestore for the full backend cutover.
- Wrote documentation for fellow engineers on the new Firebase SDKs (**Node.js** and **Python**); trained the Ops team on dashboards and **backup procedures**.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Extended backend APIs in **TypeScript** with Firebase Cloud Functions and Express.js on Heroku for a real-time stock-price PWA.
- Built scheduled jobs querying **2M+ Firestore records** to keep customer prices current from Morningstar APIs; added responsive React/Tailwind pages.

Mind2matter

Jan 2022 – Jul 2022

Software Engineer · US Remote

- Built backends for DeFi applications using **Node.js** and Web3; delivered React UIs from Figma under tight timelines, reporting to the CTO.

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead · Uganda Remote

- Architected a **containerised microservices** backend for FIDA Uganda's case-management platform on **Apache Kafka, Docker and Kubernetes**, with NATS Streaming for service messaging.
- Built a USSD legal-aid service communicating with a **Node.js** backend over **gRPC**; led a team of 4 developers.
- Delivered a paralegal database for LASPNET Uganda and a cross-platform React Native (Expo) app with Firebase push notifications.

PROJECTS

Real-time Pub/Sub Sync Pipeline

Two-way MongoDB↔Firestore sync designed for at-least-once delivery: idempotent writes, replay- and reorder-safe, running under parallel production traffic across 20+ applications.

Node.js · Google Pub/Sub · MongoDB · Firestore

Kubernetes Microservices — FIDA Uganda

Event-driven case-management backend: containerised services on Kubernetes, Kafka for events, typed gRPC boundaries between the USSD gateway and core services.

Apache Kafka · Docker · Kubernetes · gRPC · NATS Streaming

Healthcare WhatsApp Chatbot

Prescription OCR and conversational flow over the WhatsApp Business API, integrating OpenAI and Google Vision.

Node.js · OpenAI API · Google Vision · WhatsApp Business API

Agentic Research Pipeline

Multi-agent, AI-assisted evaluation pipeline with mandatory live-verification gates before any result is scored or acted on.

Node.js · Claude Code · Playwright

EDUCATION

B.Sc. Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

LINKS

Code & projects — github.com/zac-09

SKILLS

Backend: Node.js, TypeScript, Express.js, REST & gRPC APIs, Microservices, Firebase Cloud Functions

Infrastructure: Docker, Kubernetes, Linux, GCP (Firestore, Pub/Sub, Cloud Functions, GCS), AWS (S3, EC2, EBS — intermediate)

Messaging & Data: Google Pub/Sub, Apache Kafka, NATS Streaming, MongoDB, Firestore, SQL

Languages: JavaScript, TypeScript, Python, Go (intermediate), Rust (intermediate) **Spoken:** English, Kiswahili